INSTRUCTIONS

Preliminary GEK-45117A

MAIN CONTROL
193X530BBG02

ELEMENTARY DIAGRAM	36D868824AB
CARD DIAGRAM	36C764173AB

REFER TO THE DRIVE SYSTEM INSTRUCTIONS AND ELEMENTARY DIAGRAMS FOR DESCRIPTIONS, ADJUSTMENTS AND ON—CARD JUMPER CONNECTIONS.

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.

GENERAL ELECTRIC

GEK-45117A (12/79) 1M (F)

GENERAL ELECTRIC

36D868824AB

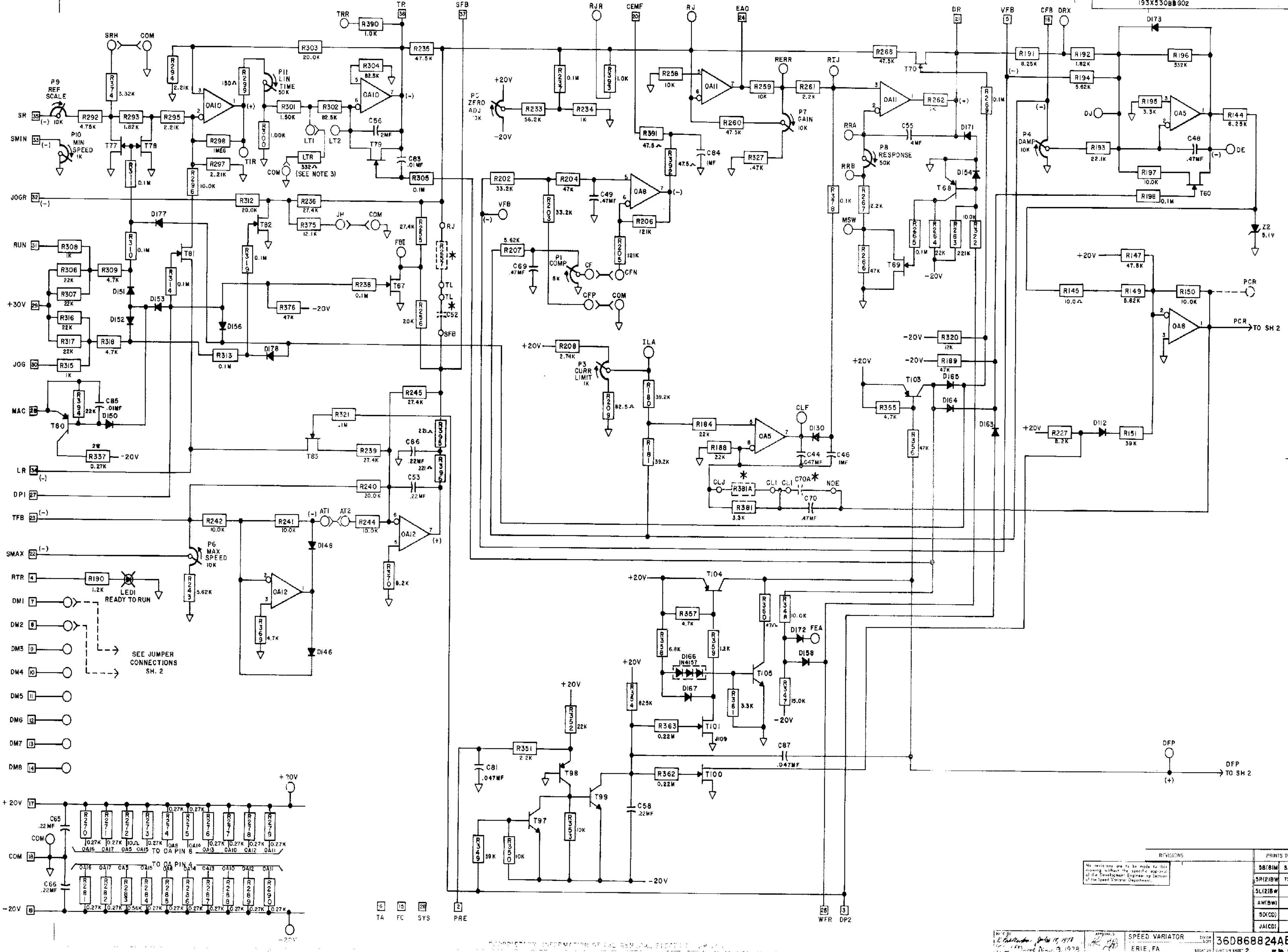
CONT ON SHEET 2 SH NO 1

ELEMENTARY DIAGRAM
DC 3060 CONTROL CARD

FIRST MADE FOR STANDARD LINE

193X530BBG02

APPLIED PRACTICES ✓



SR 35
SMIN 33
JOGR 32
RUN 31
+30V 29
JOG 30
MAC 28
LR 34
DPI 27
TFB 23
SMAX 22
RTR 4
DM1 7
DM2 8
DM3 9
DM4 10
DM5 11
DM6 12
DM7 13
DM8 14
+20V 17
COM 18
-20V 19
TA 6
FC 15
SYS 26
PRE 2
WFR 26
DP2 3
TRR
TR 36
SFB 37
RJR
CEMF 20
RJ
EAO 24
DR 21
VFB 5
CFB 16
DRX
P9 REF SCALE 10K
P10 MIN SPEED 1K
P11 LIN TIME 50K
PC ZERO ADJ 1K
P7 GAIN 10K
P8 RESPONSE 50K
P4 DAMP 10K
P3 CURR LIMIT 1K
P1 COMP 5K
P6 MAX SPEED 10K
LTR 332Ω (SEE NOTE 3)
LED1 READY TO RUN
SEE JUMPER CONNECTIONS SH. 2
OA10
OA11
OA5
OA8
OA12
T77
T78
T79
T81
T82
T67
T80
T83
T70
T68
T69
T60
T103
T104
T105
T101
T100
T98
T99
T97
Z2 5.1V
PCR
PCR TO SH 2
DFP
DFP TO SH 2
ILA
CLF
CLJ
CL1
NOE
C70A
R381A
FEA
D166 1N4157
TO OA PIN 8
TO OA PIN 4

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

PRINTS TO	
5B(8)M	5J(BW)
5R(2)BW	TS(BO)
5L(2)BW	
AW(BW)	
5D(CD)	
JA(CD)	

MADE BY K Pasternak July 19, 1978
Nov. 3, 1978

SPEED VARIATOR
ERIE, PA

36D868824AB

CONT ON SHEET 2 SH NO 1

GENERAL ELECTRIC 36D868824AB

CONT ON SHEET FL SH NO. 2

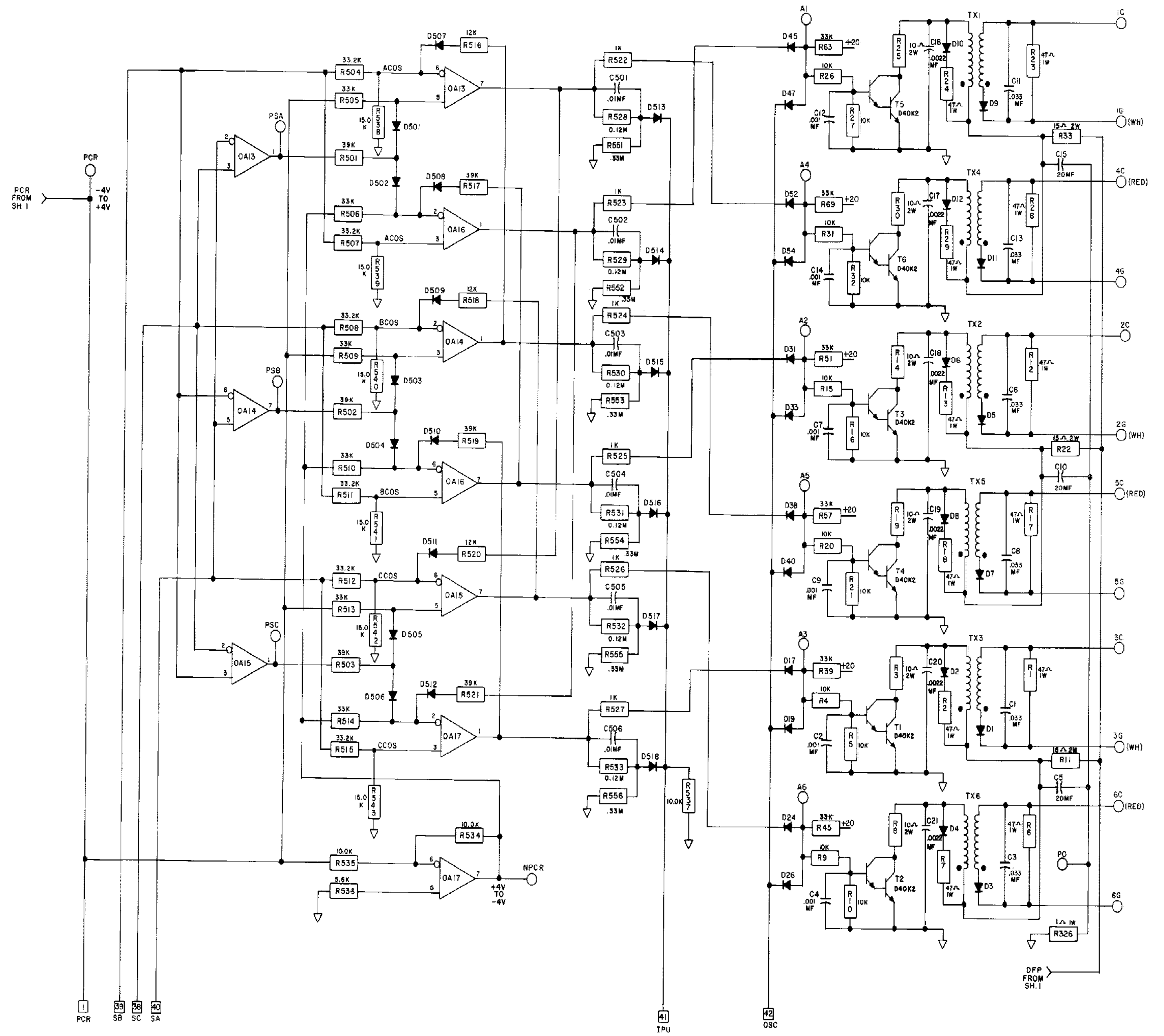
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONAL	DECIMALS	ANGLES
718A503 & 718A504	✓	+ −	± −	± −

REV NO 0

CONT ON SHEET SH NO

TITLE ELEMENTARY DIAGRAM
DC 3060 CONTROL CARD

FIRST MADE FOR STANDARD LINE
193X530BBG02



A1
A4
A2
A5
A3
A6
D45
D47
D52
D54
D31
D33
D38
D40
D17
D19
D24
D26
R63 33K +20
R26 10K
R69 33K +20
R31 10K
R51 33K +20
R15 10K
R57 33K +20
R20 10K
R39 33K +20
R4 10K
R45 33K +20
R9 10K
C12 .001 MF
C14 .001 MF
C7 .001 MF
C9 .001 MF
C2 .001 MF
C4 .001 MF
R27 10K
R32 10K
R16 10K
R21 10K
R5 10K
R10 10K
T5 D40K2
T6 D40K2
T3 D40K2
T4 D40K2
T1 D40K2
T2 D40K2
R25 10Ω 2W
R30 10Ω 2W
R14 10Ω 2W
R19 10Ω 2W
R3 10Ω 2W
R8 10Ω 2W
C16 .0022 MF
C17 .0022 MF
C18 .0022 MF
C19 .0022 MF
C20 .0022 MF
C21 .0022 MF
D10
D12
D6
D8
D2
D4
R24 47Ω 1W
R29 47Ω 1W
R13 47Ω 1W
R18 47Ω 1W
R2 47Ω 1W
R7 47Ω 1W
TX1
TX4
TX2
TX5
TX3
TX6
D9
D11
D5
D7
D1
D3
C11 .033 MF
C13 .033 MF
C6 .033 MF
C8 .033 MF
C1 .033 MF
C3 .033 MF
R23 47Ω 1W
R28 47Ω 1W
R12 47Ω
R17 47Ω 1W
R1 47Ω 1W
R6 47Ω 1W
R33 15Ω 2W
R22 15Ω 2W
R11 15Ω 2W
C15 20MF
C10 20MF
C5 20MF
R326 1Ω 1W
1C
1G (WH)
4C (RED)
4G
2C
2G (WH)
5C (RED)
5G
3C
3G (WH)
6C (RED)
6G
PO
DFP FROM SH.1
42 OSC
41 TPU
R522 1K
R523 1K
R524 1K
R525 1K
R526 1K
R527 1K
C501 .01MF
C502 .01MF
C503 .01MF
C504 .01MF
C505 .01MF
C506 .01MF
R528 0.12M
R529 0.12M
R530 0.12M
R531 0.12M
R532 0.12M
R533 0.12M
R561 .33M
R552 .33M
R553 .33M
R554 .33M
R555 .33M
R556 .33M
D513
D514
D515
D516
D517
D518
R537 10.0K
D507
D508
D509
D510
D511
D512
R516 12K
R517 39K
R518 12K
R519 39K
R520 12K
R521 39K
R504 33.2K
R505 33K
R506 33K
R507 33.2K
R508 33.2K
R509 33K
R510 33K
R511 33.2K
R512 33.2K
R513 33K
R514 33K
R515 33.2K
ACOS
BCOS
CCOS
R538 15.0K
R539 15.0K
R540 15.0K
R541 15.0K
R542 15.0K
R543 15.0K
D501
D502
D503
D504
D505
D506
R501 39K
R502 39K
R503 39K
OA13
OA16
OA14
OA16
OA15
OA17
OA13
OA14
OA15
PSA
PSB
PSC
R534 10.0K
R535 10.0K
R536 5.6K
OA17
NPCR
+4V TO -4V
PCR
-4V TO +4V
PCR FROM SH 1
1 PCR
39 SB
38 SC
40 SA

SYMBOLS

✱ – COMPONENTS SELECTED IN TEST (IF REQUIRED).

–☐ – T. B. FINGER

–○ – CONTACT PIN (FOR JUMPER CONNECTIONS AND/OR TEST POINTS)

[ ] – COMPONENTS SHOWN DOTTED ARE FURNISHED ONLY WHEN REQUIRED. C52 AND R257 WILL PROVIDE A TACHOMETER RATE NETWORK. C70A AND R381A WILL CONTROL THE RATE OF RISE OF CURRENT.

REFER TO THE INSTRUCTION BOOK FOR ADDITIONAL INFORMATION.

EXCEPT AS NOTED ALL NPN TRANSISTORS ARE 2N4424 AND ALL PNP TRANSISTORS ARE 2N5366.

JUMPER CONNECTIONS

1. THIS CARD IS SUPPLIED WITH THE JUMPERS INDICATED BELOW. OTHER JUMPER ARRANGEMENTS MAY BE MADE AS REQUIRED.

SRH — COM, JH — COM – (-10 TO -20V MAX REF)

AT1 — AT2 – (RECTIFIED TACH FEEDBACK)
CF — CFN, CFP — COM – (NEG. IR COMPENSATION)

2. DM1-ILA – ADD JUMPER FOR REMOTE CURRENT LIMIT.

DM2-FEA – ADD JUMPER FOR DELAYED FIELD ECONOMY.

3. 332-Ω RESISTOR LTR FROM LT1 TO COM FOR 2–60 SEC. LINEAR TIMING RANGE.

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

PRINTS TO	
5B(6)M	5J(BW)
5R(2)BW	T9(BO)
5L(2)BW	
AW(BW)	
5D(CD)	
JA(CD)	

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

MADE BY K. Pastemba July 19, 1971
ISSUED Paul Novak Nov. 9, 1978

APPROVALS

SPEED VARIATOR DIV OR DEPT
ERIE, PA LOCATION

36D868824AB

CONT ON SHEET FL SH NO. 2

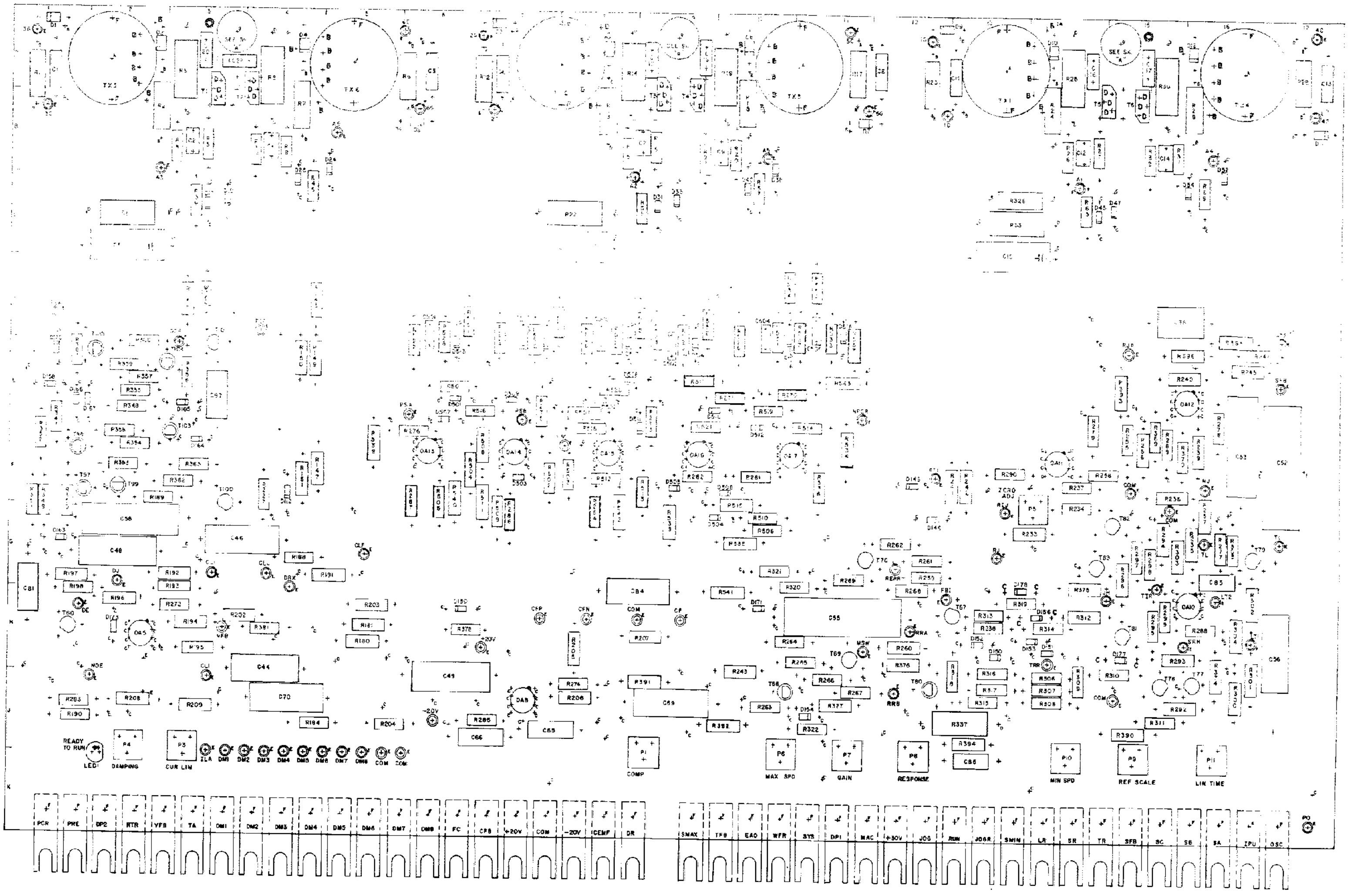
HOLE TABULATION

ALL HOLES .040 DIA EXCEPT THE HOLES TABULATED BELOW

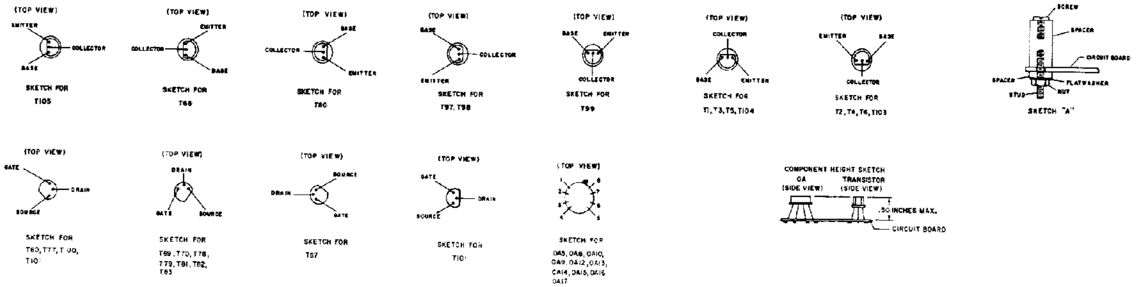
SYM.	DIA.	QUAN.
A	.157	6
B	.062	74
C	.070	220
D	.062	38
E	.070	77
F	.128	54
G	.157	1

NOTE

CARD SIZE 4.00 x 7.00


TX1
TX2
TX3
TX4
TX5
TX6
SEE SK "A"
READY TO RUN
LED1
DAMPING
CUR LIM
ZLA
DM1
DM2
DM3
DM4
DM5
DM6
DM7
DM8
COM
COMP
MAX SPD
GAIN
RESPONSE
MIN SPD
REF SCALE
LIN TIME
ZERO ADJ
PCR
PRE
DP2
RTR
VFB
TA
DM1
DM2
DM3
DM4
DM5
DM6
DM7
DM8
FC
CFB
+20V
COM
-20V
ICEMF
DR
SMAX
TFB
EAO
WFR
SYS
DP1
MAC
+30V
JOG
RUN
JOGR
SMIN
LR
SR
TR
SFB
SC
SB
SA
ZPU
OSC

TRANSISTOR LEAD SKETCHES


(TOP VIEW)
EMITTER
BASE
COLLECTOR
SKETCH FOR T105
SKETCH FOR T66
SKETCH FOR T80
SKETCH FOR T97, T98
SKETCH FOR T99
SKETCH FOR T1, T3, T5, T104
SKETCH FOR T2, T4, T6, T103
SCREW
SPACER
CIRCUIT BOARD
FLATWASHER
NUT
STUD
SKETCH "A"
GATE
DRAIN
SOURCE
SKETCH FOR T60, T77, T100, T101
SKETCH FOR T69, T70, T78, T79, T81, T82, T83
SKETCH FOR T67
SKETCH FOR T101
SKETCH FOR OA5, OA8, OA10, OA11, OA12, OA13, OA14, OA15, OA16, OA17
COMPONENT HEIGHT SKETCH
OA (SIDE VIEW)
TRANSISTOR (SIDE VIEW)
.50 INCHES MAX.
CIRCUIT BOARD

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

PRINTS TO	
5B(8)W	5J(B)W
5R(2)BW	T5(50)
5L(2)BW	
AW(B)W	
5C(CD)	
JA(CD)	

X	36C774102AB005	36C774103AB005	—	—
SCALE	REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.	FRONT	BACK
			SPACE UNIT REQUIREMENT	

UNLESS OTHERWISE SPECIFIED
DIMENSIONS ARE IN INCHES—
TOLERANCES ON
2 PLACE DECIMALS ±
3 PLACE DECIMALS ±
ANGLES ±
FRACTIONS ±
MATERIAL—

ALL SURF. ✓

SIGNATURES	DAY	MO.	YR.
DRAWN J. R. Baker	21	8	78
CHECKED			
ISSUED B. Novak	9	11	78
ENGRG.			
MFG.			
ME.			

GENERAL ELECTRIC
speed variator DEPT. LOC. ERIE, PA., U.S.A.

CONTROL CARD

COMPONENT DIAGRAM

SIZE E
CODE IDENT NO.
36C764173AB
SCALE
193X5308BG02
SHEET